```
pip install biopython
```

```
Collecting biopython
  Downloading biopython-1.87-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (13 kB)
Requirement already sat Run all cells in notebook usr/local/lib/python3.12/dist-packages (from biopython) (2.0.2)
Downloading biopython-1.87-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.2/3.2 MB 60.0 MB/s eta 0:00:00
Installing collected packages: biopython
Successfully installed biopython-1.87
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import scipy as staticmethod
```

```
pip install GEOparse
```

```
Collecting GEOparse
  Downloading GEOparse-2.0.4-py3-none-any.whl.metadata (6.5 kB)
Requirement already satisfied: numpy>=1.7 in /usr/local/lib/python3.12/dist-packages (from GEOparse) (2.0.2)
Requirement already satisfied: pandas>=0.17 in /usr/local/lib/python3.12/dist-packages (from GEOparse) (2.2.2)
Requirement already satisfied: requests>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from GEOparse) (2.32.4)
Requirement already satisfied: tqdm>=4.31.1 in /usr/local/lib/python3.12/dist-packages (from GEOparse) (4.67.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.17->GEOparse) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.17->GEOparse) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.17->GEOparse) (2026.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.21.0->GEOpa
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.21.0->GEOparse) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.21.0->GEOparse) (
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.21.0->GEOparse) (
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=0.17->G
Downloading GEOparse-2.0.4-py3-none-any.whl (29 kB)
Installing collected packages: GEOparse
Successfully installed GEOparse-2.0.4
```

```
import GEOparse
gse=GEOparse.get_GEO(geo="GSE11121")
```

```
30-Jul-2026 13:33:50 DEBUG utils - Directory ./ already exists. Skipping.
DEBUG:GEOparse:Directory ./ already exists. Skipping.
30-Jul-2026 13:33:50 INFO GEOparse - Downloading ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE11nnn/GSE11121/soft/GSE11121_family.
INFO:GEOparse:Downloading ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE11nnn/GSE11121/soft/GSE11121_family.soft.gz to ./GSE11121_f
100%|██████████| 27.9M/27.9M [00:04<00:00, 6.28MB/s]
30-Jul-2026 13:33:56 DEBUG downloader - Size validation passed
DEBUG:GEOparse:Size validation passed
30-Jul-2026 13:33:56 DEBUG downloader - Moving /tmp/tmpd06jcaxp to /content/GSE11121_family.soft.gz
DEBUG:GEOparse:Moving /tmp/tmpd06jcaxp to /content/GSE11121_family.soft.gz
30-Jul-2026 13:33:56 DEBUG downloader - Successfully downloaded ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE11nnn/GSE11121/soft/G
DEBUG:GEOparse:Successfully downloaded ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE11nnn/GSE11121/soft/GSE11121_family.soft.gz
30-Jul-2026 13:33:56 INFO GEOparse - Parsing ./GSE11121_family.soft.gz:
INFO:GEOparse:Parsing ./GSE11121_family.soft.gz:
30-Jul-2026 13:33:56 DEBUG GEOparse - DATABASE: GeoMiame
DEBUG:GEOparse:DATABASE: GeoMiame
30-Jul-2026 13:33:56 DEBUG GEOparse - SERIES: GSE11121
DEBUG:GEOparse:SERIES: GSE11121
30-Jul-2026 13:33:56 DEBUG GEOparse - PLATFORM: GPL96
DEBUG:GEOparse:PLATFORM: GPL96
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282373
DEBUG:GEOparse:SAMPLE: GSM282373
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282374
DEBUG:GEOparse:SAMPLE: GSM282374
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282375
DEBUG:GEOparse:SAMPLE: GSM282375
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282376
DEBUG:GEOparse:SAMPLE: GSM282376
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282377
DEBUG:GEOparse:SAMPLE: GSM282377
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282378
DEBUG:GEOparse:SAMPLE: GSM282378
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282379
DEBUG:GEOparse:SAMPLE: GSM282379
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282380
DEBUG:GEOparse:SAMPLE: GSM282380
30-Jul-2026 13:33:56 DEBUG GEOparse - SAMPLE: GSM282381
DEBUG:GEOparse:SAMPLE: GSM282381
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282382
DEBUG:GEOparse:SAMPLE: GSM282382
```

```
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282383
DEBUG:GEOparse:SAMPLE: GSM282383
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282384
DEBUG:GEOparse:SAMPLE: GSM282384
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282385
DEBUG:GEOparse:SAMPLE: GSM282385
30-Jul-2026 13:33:57 DE Run all cells in notebook PLE: GSM282386
DEBUG:GEOparse:SAMPLE: GSM282386
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282387
DEBUG:GEOparse:SAMPLE: GSM282387
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282388
DEBUG:GEOparse:SAMPLE: GSM282388
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282389
DEBUG:GEOparse:SAMPLE: GSM282389
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282390
DEBUG:GEOparse:SAMPLE: GSM282390
30-Jul-2026 13:33:57 DEBUG GEOparse - SAMPLE: GSM282391
DEBUG:GEOparse:SAMPLE: GSM282391
```

```
gse=GEOparse.get_GEO(geo="GSE341225",destdir=".")
```

```
30-Jul-2026 08:56:27 DEBUG utils - Directory . already exists. Skipping.
DEBUG:GEOparse:Directory . already exists. Skipping.
30-Jul-2026 08:56:27 INFO GEOparse - Downloading ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE341nnn/GSE341225/soft/GSE341225_family
INFO:GEOparse:Downloading ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE341nnn/GSE341225/soft/GSE341225_family.soft.gz to ./GSE341225
100%|██████████| 4.60k/4.60k [00:00<00:00, 10.3kB/s]
30-Jul-2026 08:56:28 DEBUG downloader - Size validation passed
DEBUG:GEOparse:Size validation passed
30-Jul-2026 08:56:28 DEBUG downloader - Moving /tmp/tmp4ynn6uik to /content/GSE341225_family.soft.gz
DEBUG:GEOparse:Moving /tmp/tmp4ynn6uik to /content/GSE341225_family.soft.gz
30-Jul-2026 08:56:28 DEBUG downloader - Successfully downloaded ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE341nnn/GSE341225/soft/G
DEBUG:GEOparse:Successfully downloaded ftp://ftp.ncbi.nlm.nih.gov/geo/series/GSE341nnn/GSE341225/soft/GSE341225_family.soft.gz
30-Jul-2026 08:56:28 INFO GEOparse - Parsing ./GSE341225_family.soft.gz:
INFO:GEOparse:Parsing ./GSE341225_family.soft.gz:
30-Jul-2026 08:56:28 DEBUG GEOparse - DATABASE: GeoMiame
DEBUG:GEOparse:DATABASE: GeoMiame
30-Jul-2026 08:56:28 DEBUG GEOparse - SERIES: GSE341225
DEBUG:GEOparse:SERIES: GSE341225
30-Jul-2026 08:56:28 DEBUG GEOparse - PLATFORM: GPL24676
DEBUG:GEOparse:PLATFORM: GPL24676
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902421
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902422
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902423
DEBUG:GEOparse:SAMPLE: GSM9902423
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902424
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902426
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902428
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902429
DEBUG:GEOparse:SAMPLE: GSM9902429
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902430
DEBUG:GEOparse:SAMPLE: GSM9902430
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902431
DEBUG:GEOparse:SAMPLE: GSM9902431
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902432
DEBUG:GEOparse:SAMPLE: GSM9902432
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902433
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902434
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30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902435
DEBUG:GEOparse:SAMPLE: GSM9902435
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902436
DEBUG:GEOparse:SAMPLE: GSM9902436
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902437
DEBUG:GEOparse:SAMPLE: GSM9902437
30-Jul-2026 08:56:28 DEBUG GEOparse - SAMPLE: GSM9902438
DEBUG:GEOparse:SAMPLE: GSM9902438
```

```
print(type(gse))
print(gse)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1844/1248580602.py in <cell line: 0>()
----> 1 print(type(gse))
      2 print(gse)

NameError: name 'gse' is not defined
```

Run all cells in notebook

```
sample=list(gse.gsms.values())[0]
print(sample.table.head())
print(sample.table.columns)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1844/3452333536.py in <cell line: 0>()
----> 1 sample=list(gse.gsms.values())[0]
      2 print(sample.table.head())
      3 print(sample.table.columns)

NameError: name 'gse' is not defined
```

```
sample.table.head()
```

	ID_REF	VALUE	ABS_CALL
0	1007_s_at	3646.0	P
1	1053_at	371.7	P
2	117_at	108.6	A
3	121_at	692.0	P
4	1255_g_at	41.9	A

```
print(len(gse.gsms))
```

```
200
```

```
for gsm_name, gsm in gse.gsms.items():
  print(gsm_name, gsm.table.shape)
```

```
GSM282373 (22283, 3)
GSM282374 (22283, 3)
GSM282375 (22283, 3)
GSM282376 (22283, 3)
GSM282377 (22283, 3)
GSM282378 (22283, 3)
GSM282379 (22283, 3)
GSM282380 (22283, 3)
GSM282381 (22283, 3)
GSM282382 (22283, 3)
GSM282383 (22283, 3)
GSM282384 (22283, 3)
GSM282385 (22283, 3)
GSM282386 (22283, 3)
GSM282387 (22283, 3)
GSM282388 (22283, 3)
GSM282389 (22283, 3)
GSM282390 (22283, 3)
GSM282391 (22283, 3)
GSM282392 (22283, 3)
GSM282393 (22283, 3)
GSM282394 (22283, 3)
GSM282395 (22283, 3)
GSM282396 (22283, 3)
GSM282397 (22283, 3)
GSM282398 (22283, 3)
GSM282399 (22283, 3)
GSM282400 (22283, 3)
GSM282401 (22283, 3)
GSM282402 (22283, 3)
GSM282403 (22283, 3)
GSM282404 (22283, 3)
GSM282405 (22283, 3)
GSM282406 (22283, 3)
GSM282407 (22283, 3)
GSM282408 (22283, 3)
```

```
GSM282409 (22283, 3)
GSM282410 (22283, 3)
GSM282411 (22283, 3)
GSM282412 (22283, 3)
GSM282413 (22283, 3)
GSM282414 (22283, 3)
GSM282415 (22283, 3)
GSM282416 (22283, 3)
GSM282417 (22283, 3)
GSM282418 (22283, 3)
GSM282419 (22283, 3)
GSM282420 (22283, 3)
GSM282421 (22283, 3)
GSM282422 (22283, 3)
GSM282423 (22283, 3)
GSM282424 (22283, 3)
GSM282425 (22283, 3)
GSM282426 (22283, 3)
GSM282427 (22283, 3)
GSM282428 (22283, 3)
GSM282429 (22283, 3)
GSM282430 (22283, 3)
```

Run all cells in notebook

```
import GEOparse
gse=GEOparse.get_GEO(geo="GSE341225")
```

```
30-Jul-2026 09:14:44 DEBUG utils - Directory ./ already exists. Skipping.
DEBUG:GEOparse:Directory ./ already exists. Skipping.
30-Jul-2026 09:14:44 INFO GEOparse - File already exist: using local version.
INFO:GEOparse:File already exist: using local version.
30-Jul-2026 09:14:44 INFO GEOparse - Parsing ./GSE341225_family.soft.gz:
INFO:GEOparse:Parsing ./GSE341225_family.soft.gz:
30-Jul-2026 09:14:44 DEBUG GEOparse - DATABASE: GeoMiame
DEBUG:GEOparse:DATABASE: GeoMiame
30-Jul-2026 09:14:44 DEBUG GEOparse - SERIES: GSE341225
DEBUG:GEOparse:SERIES: GSE341225
30-Jul-2026 09:14:44 DEBUG GEOparse - PLATFORM: GPL24676
DEBUG:GEOparse:PLATFORM: GPL24676
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902421
DEBUG:GEOparse:SAMPLE: GSM9902421
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902422
DEBUG:GEOparse:SAMPLE: GSM9902422
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902423
DEBUG:GEOparse:SAMPLE: GSM9902423
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902424
DEBUG:GEOparse:SAMPLE: GSM9902424
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902425
DEBUG:GEOparse:SAMPLE: GSM9902425
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902426
DEBUG:GEOparse:SAMPLE: GSM9902426
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902427
DEBUG:GEOparse:SAMPLE: GSM9902427
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902428
DEBUG:GEOparse:SAMPLE: GSM9902428
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902429
DEBUG:GEOparse:SAMPLE: GSM9902429
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902430
DEBUG:GEOparse:SAMPLE: GSM9902430
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902431
DEBUG:GEOparse:SAMPLE: GSM9902431
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902432
DEBUG:GEOparse:SAMPLE: GSM9902432
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902433
DEBUG:GEOparse:SAMPLE: GSM9902433
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902434
DEBUG:GEOparse:SAMPLE: GSM9902434
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902435
DEBUG:GEOparse:SAMPLE: GSM9902435
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902436
DEBUG:GEOparse:SAMPLE: GSM9902436
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902437
DEBUG:GEOparse:SAMPLE: GSM9902437
30-Jul-2026 09:14:44 DEBUG GEOparse - SAMPLE: GSM9902438
DEBUG:GEOparse:SAMPLE: GSM9902438
```

```
import GEOparse
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA
from scipy.cluster.hierarchy import linkage, dendrogram
```

```
gse=GEOparse.get_GEO(geo="GSE11121")
```

```
30-Jul-2026 13:42:31 DI Run all cells in notebook ory ./ already exists. Skipping.
DEBUG:GEOparse:Directory ./ already exists. Skipping.
30-Jul-2026 13:42:31 INFO GEOparse - File already exist: using local version.
INFO:GEOparse:File already exist: using local version.
30-Jul-2026 13:42:31 INFO GEOparse - Parsing ./GSE11121_family.soft.gz:
INFO:GEOparse:Parsing ./GSE11121_family.soft.gz:
30-Jul-2026 13:42:31 DEBUG GEOparse - DATABASE: GeoMiame
DEBUG:GEOparse:DATABASE: GeoMiame
30-Jul-2026 13:42:31 DEBUG GEOparse - SERIES: GSE11121
DEBUG:GEOparse:SERIES: GSE11121
30-Jul-2026 13:42:31 DEBUG GEOparse - PLATFORM: GPL96
DEBUG:GEOparse:PLATFORM: GPL96
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282373
DEBUG:GEOparse:SAMPLE: GSM282373
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282374
DEBUG:GEOparse:SAMPLE: GSM282374
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282375
DEBUG:GEOparse:SAMPLE: GSM282375
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282376
DEBUG:GEOparse:SAMPLE: GSM282376
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282377
DEBUG:GEOparse:SAMPLE: GSM282377
30-Jul-2026 13:42:31 DEBUG GEOparse - SAMPLE: GSM282378
DEBUG:GEOparse:SAMPLE: GSM282378
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282379
DEBUG:GEOparse:SAMPLE: GSM282379
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282380
DEBUG:GEOparse:SAMPLE: GSM282380
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282381
DEBUG:GEOparse:SAMPLE: GSM282381
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282382
DEBUG:GEOparse:SAMPLE: GSM282382
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282383
DEBUG:GEOparse:SAMPLE: GSM282383
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282384
DEBUG:GEOparse:SAMPLE: GSM282384
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282385
DEBUG:GEOparse:SAMPLE: GSM282385
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282386
DEBUG:GEOparse:SAMPLE: GSM282386
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282387
DEBUG:GEOparse:SAMPLE: GSM282387
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282388
DEBUG:GEOparse:SAMPLE: GSM282388
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282389
DEBUG:GEOparse:SAMPLE: GSM282389
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282390
DEBUG:GEOparse:SAMPLE: GSM282390
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282391
DEBUG:GEOparse:SAMPLE: GSM282391
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282392
DEBUG:GEOparse:SAMPLE: GSM282392
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282393
DEBUG:GEOparse:SAMPLE: GSM282393
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282394
DEBUG:GEOparse:SAMPLE: GSM282394
30-Jul-2026 13:42:32 DEBUG GEOparse - SAMPLE: GSM282395
DEBUG:GEOparse:SAMPLE: GSM282395
```

```
expression=gse.pivot_samples("VALUE")
print("Expression Matrix")
print(expression.head())
print("Shape:",expression.shape)
```

```
Expression Matrix
name       GSM282373  GSM282374  GSM282375  GSM282376  GSM282377  GSM282378  \
ID_REF
1007_s_at     3646.0     6200.0     3529.0     4200.0     3368.0     1753.0
1053_at        371.7      336.2      321.6      432.3      461.5      552.1
117_at         108.6      173.6      148.7      145.1      135.1      137.1
121_at         692.0      641.1      524.5      783.6      862.6      558.2
1255_g_at       41.9       40.7       20.1       43.8       58.3       15.6

name       GSM282379  GSM282380  GSM282381  GSM282382  ...  GSM282563  \
ID_REF                                                 ...
1007_s_at     4042.0     2526.0     2934.0     2824.0  ...     2438.0
1053_at        231.8      399.4      343.4      237.0  ...      252.1
```

```
117_at         141.2      150.9      127.7       15.8  ...      172.3
121_at         450.2      445.6      688.9      525.5  ...      450.6
1255_g_at       33.5       14.3       28.7       26.8  ...        9.4

name       GSM282564  GSM282565  GSM282566  GSM282567  GSM282568  GSM282569  \
ID_REF
1007_s_at     2261.0     2...       ...        3336.0     3834.0     2503.0
1053_at        238.7      505.9      293.3      398.1      875.7      681.8
117_at         118.5      176.7       97.2      198.0      345.1      164.8
121_at         504.1      523.5      481.5      551.2      536.7      810.6
1255_g_at       35.9        6.0        6.2       41.4       11.3       24.1

name       GSM282570  GSM282571  GSM282572
ID_REF
1007_s_at     2288.0     2471.0     5407.0
1053_at        238.6      198.5      252.8
117_at         338.6      257.2      233.1
121_at         770.3      778.3      918.5
1255_g_at        8.4       50.9       11.3

[5 rows x 200 columns]
Shape: (22283, 200)
```

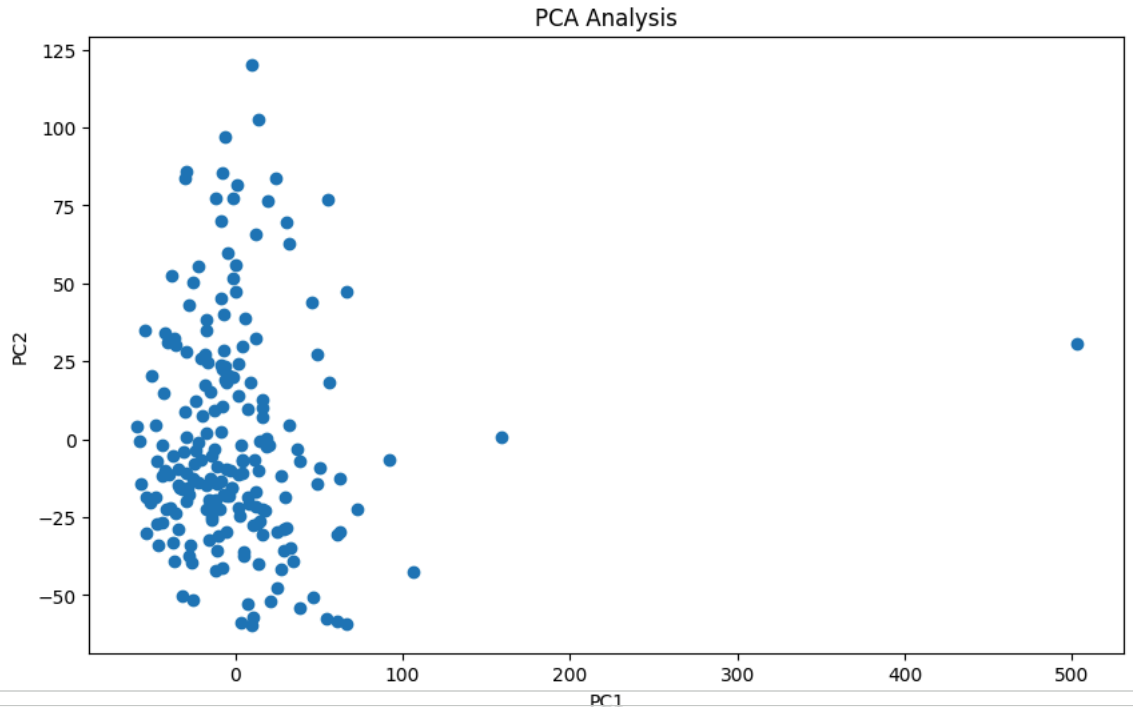
Run all cells in notebook

```
X=expression.T
scaler=StandardScaler()
X_scaled=scaler.fit_transform(X)
```

```
pca=PCA(n_components=2)
X_pca=pca.fit_transform(X_scaled)
pca_df=pd.DataFrame(data=X_pca,columns=["PC1","PC2"])
print(pca_df.head())
```

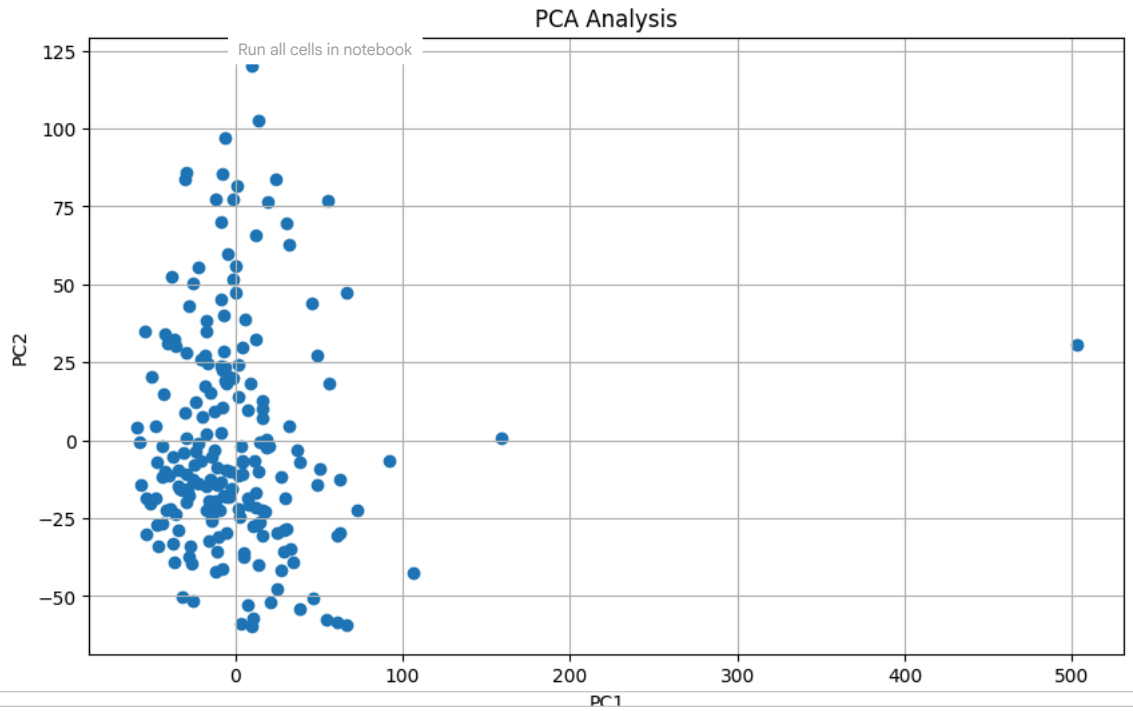
```
         PC1        PC2
0  12.724187  32.331818
1 -13.678404 -25.668477
2 -14.074310 -24.764627
3  12.197043 -16.853074
4   2.217051 -22.156088
```

```
plt.figure(figsize=(10,6))
plt.scatter(pca_df["PC1"],pca_df["PC2"])
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA Analysis")
plt.show()
```

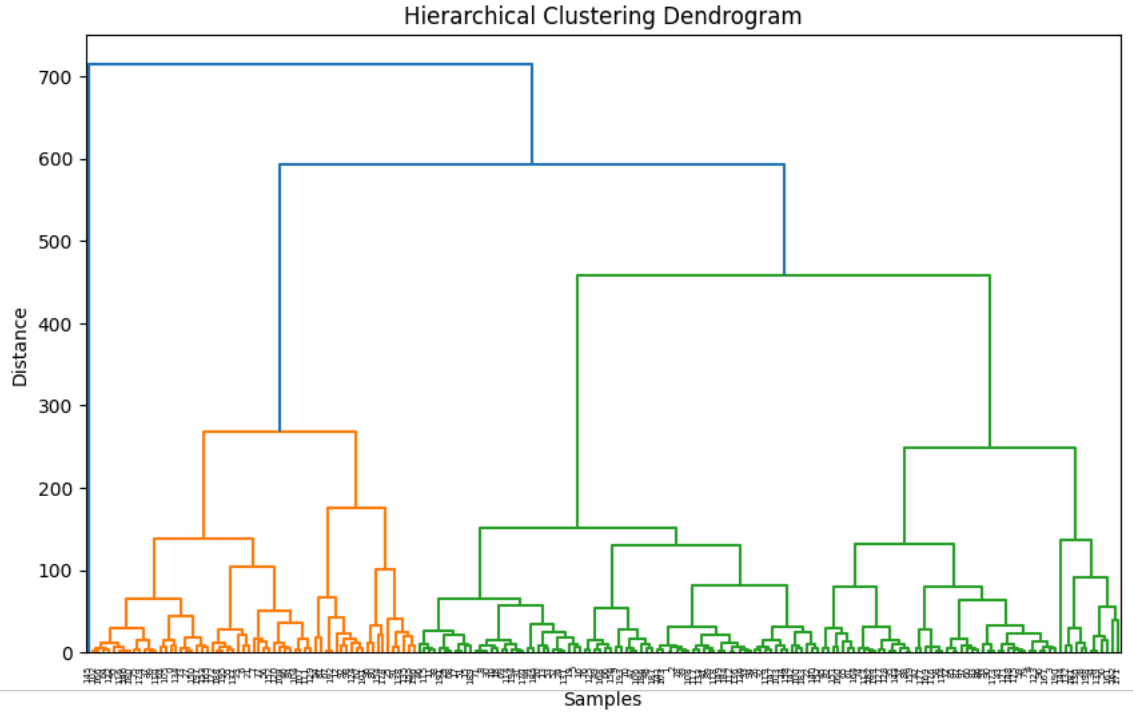



```
plt.figure(figsize=(10,6))
plt.scatter(pca_df["PC1"],pca_df["PC2"])
plt.xlabel("PC1")
```

```
plt.ylabel("PC2")
plt.title("PCA Analysis")
plt.grid(True)
```




```
linked=linkage(X_pca,method="ward")
plt.figure(figsize=(10,6))
dendrogram(linked)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()
```



Double-click (or enter) to edit

```
expression.to_csv("GSE11121_expression_matrix.csv")
```

```
gse=GEOparse.get_GEO(geo="GSE11121")
for gsm, sample in gse.gsms.items():
```

```
  print(gsm)
  print(sample.metadata.get("Characteristics_ch1"))
  print("-"*50)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1844/33{ Run all cells in notebook l line: 0>()
----> 1 for gsm, sample in gse.gsms.items():
      2   print(gsm)
      3   print(sample.metadata.get("Characteristics_ch1"))
      4   print("-"*50)

NameError: name 'gse' is not defined
```

```
sample=list(gse.gsms.values())[0]
print(sample.metadata.keys())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1844/1487778735.py in <cell line: 0>()
----> 1 sample=list(gse.gsms.values())[0]
      2 print(sample.metadata.keys())

NameError: name 'gse' is not defined
```

```
for key,value in sample.metadata.items():
  print(key,value)
```

```
title ['BC6001_r: Mainz_N0_untreated_frozen _001']
geo_accession ['GSM282373']
status ['Public on Jul 01 2008']
submission_date ['Apr 17 2008']
last_update_date ['Sep 21 2020']
type ['RNA']
channel_count ['1']
source_name_ch1 ['primary breast cancer tissue']
organism_ch1 ['Homo sapiens']
taxid_ch1 ['9606']
characteristics_ch1 ['storage: fresh frozen', 'node: 0', 'grade: 2', 'size_in_cm: 1.8', 't.dmfs: 92', 'e.dmfs: 1', 'clinic: Main
treatment_protocol_ch1 ['For all tumors, samples were snap frozen and stored at -80°C. Tumor cell content exceeded 40% in all sa
molecule_ch1 ['total RNA']
extract_protocol_ch1 ["Approximately 50 mg of frozen breast tumor tissue was crushed in liquid nitrogen. RLT-Buffer was added an
label_ch1 ['biotin']
label_protocol_ch1 ['Starting from 5 µg total RNA, labeled cRNA was prepared using the Roche Microarray cDNA Synthesis, Microarr
hyb_protocol ['Following fragmentation, 15 ug of cRNA were hybridized for 16 hr at 45C on GeneChip HG-U133a Array. GeneChips wer
scan_protocol ['GeneChips were scanned using the Hewlett-Packard GeneArray Scanner G2500A.']
description ['Gene expression data from primary breast tumors']
data_processing ['The data were analyzed with Microarray Suite version 5.0 (MAS 5.0) using Affymetrix default analysis settings
platform_id ['GPL96']
contact_name ['Mathias,,Gehrmann']
contact_email ['mathias.gehrmann@bayer.com']
contact_phone ['+49-214-30-49539']
contact_fax ['+49-214-30-50698']
contact_department ['BTS-PT-AS-AT']
contact_institute ['Bayer Technology Services GmbH']
contact_address ['Kaiser-Wilhelm-Allee']
contact_city ['Leverkusen']
contact_zip/postal_code ['51368']
contact_country ['Germany']
supplementary_file ['ftp://ftp.ncbi.nlm.nih.gov/geo/samples/GSM282nnn/GSM282373/suppl/GSM282373.CEL.gz']
relation ['Reanalyzed by: GSE47561', 'Reanalyzed by: GSM4797012']
series_id ['GSE11121']
data_row_count ['22283']
```

```
import matplotlib.pyplot as plt
import numpy as np
results["neg_log10_pvalue"]=-np.log10(results["pvalue"])
plt.figure(figsize=(8,6))
```

Run all cells in notebook